

SECURITY OF QUANTUM SECRET SHARING WITH TWO-PARTICLE ENTANGLEMENT AGAINST INDIVIDUAL ATTACKS

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Security is the most important criterion to evaluate cryptography protocols. We investigate the security of one important quantum secret sharing protocol [Phys. Rev. A 59, 162 (1999)] (KKI protocol). By discriminating two mixed states, we derive the optimal relationship between the induced error rate (QBER) and the maximal amount of information gained by a dishonest participant. For a specific amount of eavesdropper's knowledge on the shared secret that we consider, the disturbance on the quantum state in terms of QBER is actually larger for the hybrid protocol than the KKI protocol. Therefore, the hybrid protocol appears to be better able to detect eavesdroppers than the KKI protocol. For this reason, we do not see any reason to employ the KKI protocol against individual attacks.

Keywords: quantum secret sharing, quantum cryptography, individual attack, security

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Quantum cryptography, as first suggested by Bennett and Brassard (BB84) [1], is the most advanced development of quantum information processing. Secret sharing is a well-studied field of cryptography which allows a secret to be shared among many participants in such a way that only the authorized groups can reconstruct it. Some interesting publications have generalized secret sharing into quantum scenario. Up to now, there are two different types of quantum secret sharing (QSS). The first type is the sharing of *quantum* secrets, as first proposed by Cleve *et al.* [2]. The second type is the quantum sharing of *classical* secrets, as first studied by Hillery *et al.* [3]. Since the above pioneering works appeared, QSS has attracted a great deal of attention [4, 5, 6]. Here we focus our attention on the latter (hereafter “QSS” denotes specially the quantum sharing of classical secrets).

Karlsson *et al.* [6] proposed a novel quantum protocol to share classical keys in 1999 (in the following called KKI). It could lead to interesting experiments from a physics point of view. But, in our opinion, it is not serious secret-sharing from a cryptography perspective.

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In fact, it combined the quantum key distribution (QKD) and secret-sharing to transmit the shares securely in the presence of eavesdroppers. The original intention of KKI is that it is more efficient than the hybrid scheme, i.e. QKD followed by classical secret sharing [7, 8]. However, As we know, security is the most important requirement for a cryptographical protocol. Consequently we wondered, from the aspect of security, which way to achieve this task is better.

In this paper, we analyze the security of KKI [6] by a general individual eavesdropping strategy. To evaluate the security extent, we derive the optimal trade-off between the information gain and the error rate introduced by an eavesdropper. Furthermore, comparing with the hybrid scheme, we find that the information gained by an eavesdropper is larger for a specific error rate in KKI protocol. Therefore, the hybrid scheme is a securer way than KKI to achieve secret sharing against individual attacks.

We first introduce the process of KKI protocol. Similar with the BB84 protocol [1], the dealer Trent encodes his information with two nonorthogonal sets $\{|\psi^+\rangle, |\phi^-\rangle\}$ and $\{|\Psi^+\rangle, |\Phi^-\rangle\}$ randomly, where $|\psi^+\rangle, |\phi^-\rangle$ are two Bell states, and $|\Psi^+\rangle = (|\phi^-\rangle + |\psi^+\rangle)/\sqrt{2}$, $|\Phi^-\rangle = (|\phi^-\rangle - |\psi^+\rangle)/\sqrt{2}$. Then he sends the two qubits to his two agents Alice and Bob respectively, who perform measurements in z or x basis at random. After classical communication via public channel, Alice and Bob discard the results when their measurement bases are inconsistent. In this way, Alice and Bob can read out Trent's secret key jointly according to their remaining results.

Now let us derive the eavesdropping strategy which is optimal with respect to relationship between the information gain and the induced error rate for a eavesdropper. We do not consider collective or coherent attacks, but only the individual attack. As pointed out in Refs. [6, 9, 10, 11], a dishonest participant generally has more advantages to attack than an outside eavesdropper. If a QSS protocol is secure for the dishonest participants, it is secure for any eavesdropper. Therefore we concentrate our attention on the participant attacks, and without loss of generality, we assume that the dishonest participant is Bob, who has unlimited ability of computation and can take any eavesdropping operation allowed by the laws of quantum mechanics. To obtain Trent's secret by himself, Bob can perform a general individual attack as described in Ref. [13, 14, 15, 16, 17]. When two qubits are sent out by Trent, Bob lets a probe, initially in some standard state $|\chi\rangle$, interact unitarily with them. After the interaction, Bob sends one qubit to Alice, stores another qubit and his probe to glean Trent's information later. The most general unitary transformation Bob can design is of the form

$$\begin{aligned} U|\phi^-\rangle|\chi\rangle &= |z^+\rangle|\varepsilon_{00}\rangle + |z^-\rangle|\varepsilon_{01}\rangle, \\ U|\psi^+\rangle|\chi\rangle &= |z^+\rangle|\varepsilon_{10}\rangle + |z^-\rangle|\varepsilon_{11}\rangle. \end{aligned} \quad (1)$$

Here $|z^+\rangle = |0\rangle, |z^-\rangle = |1\rangle$. The first qubit is the one of Alice, and others are held by Bob. $|\varepsilon_{ij}\rangle, i, j \in \{0, 1\}$ refer to the states at Bob's site after the interaction, which are in general neither normalized nor orthogonal. Unitarity of the operation U implies:

$$\begin{aligned} \langle\varepsilon_{00}|\varepsilon_{10}\rangle + \langle\varepsilon_{01}|\varepsilon_{11}\rangle &= 0, \\ \langle\varepsilon_{00}|\varepsilon_{00}\rangle + \langle\varepsilon_{01}|\varepsilon_{01}\rangle &= 1, \\ \langle\varepsilon_{10}|\varepsilon_{10}\rangle + \langle\varepsilon_{11}|\varepsilon_{11}\rangle &= 1. \end{aligned} \quad (2)$$

Similar to Refs. [14, 15, 16, 17], Bob is wise enough to treat all four states in the same way, that is, Bob's eavesdropping has to be done carefully so that the honest participant (Alice) gets each outcome in equal probability for every initial state; otherwise Trent and Alice could find out eavesdropper's existence with refined data analysis (i.e., Trent compares the probabilities Alice gets each outcome according to different initial states). For example, for the initial state $U|\phi^-\rangle|\chi\rangle = |z^+\rangle|\varepsilon_{00}\rangle + |z^-\rangle|\varepsilon_{01}\rangle = \frac{1}{\sqrt{2}}|x^+\rangle(|\varepsilon_{00}\rangle + |\varepsilon_{01}\rangle) + \frac{1}{\sqrt{2}}|x^-\rangle(|\varepsilon_{00}\rangle - |\varepsilon_{01}\rangle)$, ($|x^+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|x^-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$) no matter which basis Alice chooses to measure her qubit, the probability for getting each measurement outcome must be equal, that is

$$\begin{aligned}\langle\varepsilon_{00}|\varepsilon_{00}\rangle &= \langle\varepsilon_{01}|\varepsilon_{01}\rangle = \frac{1}{2}, \\ (\langle\varepsilon_{00}|\varepsilon_{01}\rangle + \langle\varepsilon_{01}|\varepsilon_{00}\rangle) &= 0, \\ (\langle\varepsilon_{00}|\varepsilon_{01}\rangle - \langle\varepsilon_{01}|\varepsilon_{00}\rangle) &= 0.\end{aligned}\tag{3}$$

Similarly, let the probability of each outcome for other three initial states be equal, we get some constrains:

$$\begin{aligned}\langle\varepsilon_{00}|\varepsilon_{00}\rangle &= \langle\varepsilon_{01}|\varepsilon_{01}\rangle = \langle\varepsilon_{10}|\varepsilon_{10}\rangle = \langle\varepsilon_{11}|\varepsilon_{11}\rangle = \frac{1}{2}, \\ \langle\varepsilon_{00}|\varepsilon_{11}\rangle + \langle\varepsilon_{01}|\varepsilon_{10}\rangle + \langle\varepsilon_{10}|\varepsilon_{01}\rangle + \langle\varepsilon_{11}|\varepsilon_{00}\rangle &= 0, \\ \langle\varepsilon_{00}|\varepsilon_{01}\rangle + \langle\varepsilon_{01}|\varepsilon_{00}\rangle &= 0, \\ \langle\varepsilon_{00}|\varepsilon_{10}\rangle + \langle\varepsilon_{10}|\varepsilon_{00}\rangle &= 0, \\ \langle\varepsilon_{10}|\varepsilon_{11}\rangle + \langle\varepsilon_{11}|\varepsilon_{10}\rangle &= 0, \\ \langle\varepsilon_{01}|\varepsilon_{11}\rangle + \langle\varepsilon_{11}|\varepsilon_{01}\rangle &= 0.\end{aligned}\tag{4}$$

For the same reason as described in Ref. [15], we assume all the inner products are real. Therefore, the constrains can be reduced to

$$\begin{aligned}\langle\varepsilon_{00}|\varepsilon_{00}\rangle &= \langle\varepsilon_{01}|\varepsilon_{01}\rangle = \langle\varepsilon_{10}|\varepsilon_{10}\rangle = \langle\varepsilon_{11}|\varepsilon_{11}\rangle = \frac{1}{2}, \\ \langle\varepsilon_{00}|\varepsilon_{01}\rangle &= \langle\varepsilon_{00}|\varepsilon_{10}\rangle = \langle\varepsilon_{10}|\varepsilon_{11}\rangle = \langle\varepsilon_{01}|\varepsilon_{11}\rangle = 0, \\ \langle\varepsilon_{00}|\varepsilon_{11}\rangle + \langle\varepsilon_{01}|\varepsilon_{10}\rangle &= 0.\end{aligned}\tag{5}$$

It is from the states $|\varepsilon_{ij}\rangle$ and their linear combinations that Bob will glean information. Note that, irrespective of the choice of U , there can be no more than four linearly independent vectors $|\varepsilon_{ij}\rangle$. That is to say, the four vectors $|\varepsilon_{ij}\rangle$ span, at most, a four-dimensional space. Without loss of generality, we can choose basis vectors $|e_m\rangle$, $m \in \{0, 1, 2, 3\}$ to represent $|\varepsilon_{ij}\rangle$ as

$$\begin{aligned}|\varepsilon_{00}\rangle &= \frac{1}{\sqrt{2}}|e_0\rangle, \\ |\varepsilon_{01}\rangle &= \frac{1}{\sqrt{2}}|e_1\rangle, \\ |\varepsilon_{10}\rangle &= \frac{1}{\sqrt{2}}(a|e_1\rangle + b|e_2\rangle + c|e_3\rangle), \\ |\varepsilon_{11}\rangle &= \frac{1}{\sqrt{2}}(-a|e_0\rangle + c|e_2\rangle - b|e_3\rangle).\end{aligned}\tag{6}$$

where a, b, c are real and $a^2 + b^2 + c^2 = 1$.

Similar to the optimal eavesdropping of BB84 protocol [14], in order to take advantage of the delayed public information, Bob's optimal strategy is measuring his qubits after Trent and Alice reveal their classical information. However, according to the KKI protocol, in the process of eavesdropping detection, Bob and Alice must first announce his measurement outcomes for test qubits and then the measurement bases (MBs) for both test qubits and information qubits. Therefore, in this condition, the most favorable manipulations for Bob are as follows: (i) for the test qubits, Bob must announce his outcomes before Alice and Trent's announcement, thereby no "delayed information" can be used in his measurements. To reduce the probability of being detected, Bob has to perform certain measurements and give an advantageous announcement according to his measurement outcomes (Note that the measurements are not restricted in " z " or " x " bases as required in the KKI protocol. We will explain them below.); (ii) for the information qubits, to make use of the delayed information of Alice's MBs and Trent's sets, Bob announces a random sequence of z and x to forge his MBs (i.e., these MBs are fictitious, and actually he has not measured any of them and even not decided how to measure each information qubit).

Here we explain Bob's operation on the test qubits in detail. It is easy to see that Bob's eavesdropping may disturb the state of test qubits. To reduce the probability being detected, Bob chooses two optimal measurements Z' and X' randomly on the test qubits substituting for z and x respectively, which introduce the minimum error rate in each case. We care about the minimum error rate introduced by Bob instead of the measurement operations. In fact, for two states to discriminate, the minimum error rate is determinate, which only depends on the two states discriminated, and is irrelevant to the measurement. Here, we assume Z' and X' are the optimal measurements where introducing minimum error rates in two cases, respectively. Now we explore the minimum error rate inflicted on the test qubits by Bob in each case.

First consider the case that Bob performs Z' measurement, which can help Bob to give a correct announcement with maximal probability. He first announces his outcome 0 or 1 (correspond to $|z^+\rangle$ or $|z^-\rangle$) and then his MB " z ". If their results can be used to detect eavesdropping, Bob's, Alice's MBs and Trent's states must satisfy the correlation pointed out in Ref. [6]. That is when the initial state sent by Trent is $|\phi^-\rangle$ or $|\psi^+\rangle$, Alice's MB must be z , on the contrary if Trent's state is $|\Phi^-\rangle$ or $|\Psi^+\rangle$, Alice's MB is x ; otherwise their results have no determinate relation and will be discarded. Therefore Bob knows the possible states at his site when he declares z basis. For example, if the initial state is $U|\phi^-\rangle|\chi\rangle = |0\rangle|\varepsilon_{00}\rangle + |1\rangle|\varepsilon_{01}\rangle$ (Alice's MB is z), Bob's state may be $|\varepsilon_{00}\rangle$ or $|\varepsilon_{01}\rangle$ which should correspond to two outcomes $|z^+\rangle$ and $|z^-\rangle$ respectively when no error occurs. As the initial states sent by Trent is one of the four states with equal probability, the states at Bob's site may be one of the following eight states with equal probability.

$$\begin{aligned} |\varphi_1\rangle &= |e_0\rangle, \\ |\varphi_2\rangle &= -a|e_0\rangle + c|e_2\rangle - b|e_3\rangle, \\ |\varphi_3\rangle &= \frac{1}{2}[(1-a)|e_0\rangle - (1+a)|e_1\rangle - (b-c)|e_2\rangle - (c+b)|e_3\rangle], \\ |\varphi_4\rangle &= \frac{1}{2}[(1-a)|e_0\rangle + (1+a)|e_1\rangle + (b+c)|e_2\rangle + (c-b)|e_3\rangle], \end{aligned}$$

$$\begin{aligned}
|\varphi_5\rangle &= |e_1\rangle, \\
|\varphi_6\rangle &= a|e_1\rangle + b|e_2\rangle + c|e_3\rangle, \\
|\varphi_7\rangle &= \frac{1}{2}[(1+a)|e_0\rangle + (1-a)|e_1\rangle - (b+c)|e_2\rangle - (c-b)|e_3\rangle], \\
|\varphi_8\rangle &= \frac{1}{2}[(1+a)|e_0\rangle - (1-a)|e_1\rangle + (b-c)|e_2\rangle + (c+b)|e_3\rangle].
\end{aligned} \tag{7}$$

The first four states imply that Bob should declare $|z^+\rangle$, and the other four states imply $|z^-\rangle$. Therefore to judge which announcement ($|z^+\rangle$ or $|z^-\rangle$) to be given, Bob should distinguish the two mixed states ρ_{z^+} , ρ_{z^-} which occur with equal probability,

$$\rho_{z^+} = \frac{1}{4}[|\varphi_1\rangle\langle\varphi_1| + |\varphi_2\rangle\langle\varphi_2| + |\varphi_3\rangle\langle\varphi_3| + |\varphi_4\rangle\langle\varphi_4|], \tag{8}$$

$$\rho_{z^-} = \frac{1}{4}[|\varphi_5\rangle\langle\varphi_5| + |\varphi_6\rangle\langle\varphi_6| + |\varphi_7\rangle\langle\varphi_7| + |\varphi_8\rangle\langle\varphi_8|]. \tag{9}$$

According to the well-known result [12], to discriminate two mixed states ρ_0 and ρ_1 occurring with a priori probabilities p_0 and p_1 , the minimum-error probability attainable is $P_E = \frac{1}{2} - \frac{1}{2}\|p_0\rho_0 - p_1\rho_1\|$, where $\|\Lambda\| = \text{tr}\sqrt{\Lambda^\dagger\Lambda}$. After fussy calculating, we get the minimum error rate $\frac{1}{4} + \frac{a}{4}$ for Bob distinguishing his outcomes $|z^+\rangle$ and $|z^-\rangle$.

For the case that Bob performs X' measurement, using the same method, we get the minimum error probability Bob distinguish his outcomes $|x^+\rangle$ and $|x^-\rangle$ which is same as that in distinguishing $|z^+\rangle$ and $|z^-\rangle$. Therefore the average error rate introduced by Bob's operations for all possible states is

$$D = \frac{1}{4} + \frac{a}{4}. \tag{10}$$

Note that $-1 \leq a \leq 1$, and thus $0 \leq D \leq \frac{1}{2}$.

We now consider the information qubits, and derive the maximum mutual information I^{TB} (Shannon information) between Trent and Bob. As mentioned above, when Trent requires Bob to publish his MBs for information qubits, Bob declares a sequence of fictitious MBs, actually he has not measure any qubit at that time. However Bob can know Alice's MBs and Trent's sets from their public information, according to which he decides how to measure his information qubit. Now we calculate the maximal information Bob can obtain from his measurements each case.

First consider the case that Trent revealed his set was $\{|\psi^+\rangle, |\phi^-\rangle\}$, and Alice measured her qubit in z basis, Bob's states may be one of the four states with equal probability:

$$\begin{aligned}
|\omega_1\rangle &= |e_0\rangle, \\
|\omega_2\rangle &= |e_1\rangle, \\
|\omega_3\rangle &= a|e_1\rangle + b|e_2\rangle + c|e_3\rangle, \\
|\omega_4\rangle &= -a|e_0\rangle + c|e_2\rangle - b|e_3\rangle.
\end{aligned} \tag{11}$$

The first two states are from the initial states $|\phi^-\rangle$, and others are from $|\psi^+\rangle$. To deduce the initial state $|\phi^-\rangle$ or $|\psi^+\rangle$ (correspond to secret bit 0 or 1), Bob must perform a measurement to discriminate two mixed states $\rho_{\phi^-} = \frac{1}{2}(|\omega_1\rangle\langle\omega_1| + |\omega_2\rangle\langle\omega_2|)$ and $\rho_{\psi^+} = \frac{1}{2}(|\omega_3\rangle\langle\omega_3| + |\omega_4\rangle\langle\omega_4|)$. Then the minimal error rate to deduce Trent's secret is

$$Q_{min} = \frac{1}{2} - \frac{1}{2}\sqrt{1-a^2} \tag{12}$$

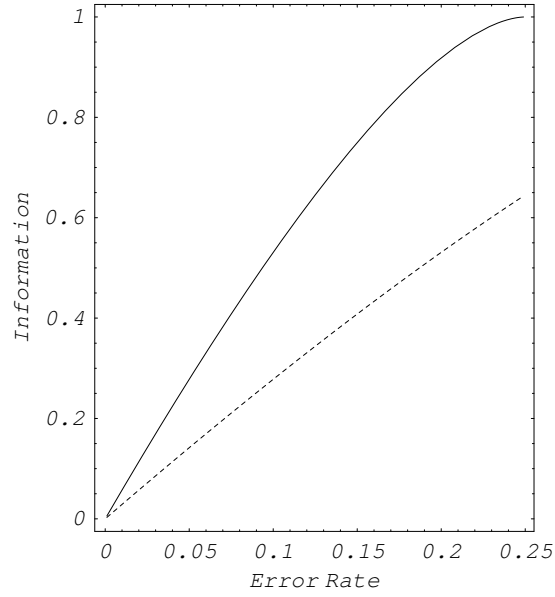


Fig. 1. The eavesdropper's information I as functions of error rate D . The lower curve holds for the hybrid scheme and is shown for the purpose of comparison.

The mutual information between Trent and Bob is given by

$$I^{TB} = 1 + Q \log Q + (1 - Q) \log(1 - Q), \quad (13)$$

where Q is the error rate to discriminate Trent's secret in each case.

For the other three cases, we get the same result as Eq. (12). Therefore the maximal average information Bob can gain is

$$I^{TB} = 1 + \frac{1 - \sqrt{1 - a^2}}{2} \log \frac{1 - \sqrt{1 - a^2}}{2} + \frac{1 + \sqrt{1 - a^2}}{2} \log \frac{1 + \sqrt{1 - a^2}}{2}. \quad (14)$$

The relationship $I(D)$ between the information gain and the induced error rate implied by Eqs.(10) and (14) is plotted in Fig.1, where we also give the corresponding relationship of the hybrid scheme, for the purpose of comparison. The hybrid scheme is another approach (QKD followed by conventional techniques) to achieve the same object as that of KKI. The solution of the hybrid scheme is as follows: the dealer Trent first establishes two random keys K_1 , K_2 using BB84 protocol, where K_1 for Alice, and K_2 for Bob. Then the secret key of Trent, is $K = K_1 + K_2$. When Trent wants to send secret message M , he can encrypt it with his random key K . In this way, Alice and Bob can decrypt Trent's secret only when they cooperate. If a dishonest participant, Bob, wants to filch Trent's secret, the only way is eavesdropping the key K_1 between Trent and Alice. His eavesdropping will induce disturbance inevitably, and the optimal trade-off [14] between information and error rate is indicated by the dashed line in Fig.1. Note that our curve lies everywhere up the one for the hybrid scheme, that is, the eavesdropper gains more information for the same error rate as that of the hybrid scheme. In

other words, the eavesdropper will introduce less noise in order to gain the same information as that of the hybrid scheme. For example, when Bob gains all the information of Trent ($I = 1$), he will introduce $\frac{1}{2}$ error rate in the hybrid scheme, but $\frac{1}{4}$ in the KKI protocol which is much easier to be concealed by channel noise. Therefore, the hybrid protocol is better able to detect eavesdroppers than the KKI protocol. For this reason, we do not see any reason to employ the KKI protocol against individual attacks.

One may argue that the comparison is unfair, as the disturbances in the hybrid scheme and KKI are computed in different ways. Here we emphasize that, the two ways to measure the disturbance are consistent. Both of them take the measure criterion in Ref. [13], that is, the disturbance is the probability that a discrepancy would be detected by participants (or error rate). In the former, the measurements to find discrepancy are determinate. That is, when the input state is $|z^+\rangle$ or $|z^-\rangle$ ($|x^+\rangle$ or $|x^-\rangle$), the measurement basis is z (x). Therefore, the error rate can be represented as a function of fidelity between input state and output state. The reason without discussing the optimal measurement for Eve is that Eve's measurement can not change the discrepancy rate detectable as she does not participate in the process of checking eavesdropping. While in the latter, the eavesdropper, Bob, is a participant in the process of eavesdropping detection. As the discrepancy detectable are the inconsistent outcomes among three parties, Bob can reduce the inconsistent outcomes by performing optimal measurements and telling a lie. Therefore, the discrepancy rate can not be represented by fidelity between input state and output state directly. Here we compute the minimum discrepancy rate using minimum error discrimination. Note that the minimum error rate for discriminating two states only depends on the two input states. Anyway, both two analyses compute the minimum discrepancy rate induced by an eavesdropper, and therefore the comparison is fair.

It should be pointed out that our analysis is more complex than that in Ref. [14], as the eavesdropper in KKI is a participant who can reduce the error rate by optimal measurements. Just because of eavesdropper's participation in process of eavesdropping detection, the security of KKI is weaker against individual attacks than the hybrid scheme. In our opinion, the security extent of QSS protocols will not exceed that of the corresponding hybrid scheme, which is indeed illustrated by our result. Therefore the study on QSS can be transferred into the study of QKD and classical secret sharing.

In conclusion, we have analyzed the security of one prominent quantum secret sharing protocol KKI, and derived the optimal trade-off between the information gain and the error rate introduced by a dishonest participant. Comparing the trade-off under the same assumptions, we found the eavesdropper obtained more information in KKI than in the hybrid scenario for a fixed error rate. Therefore the KKI protocol offers weaker security than the hybrid scheme against individual attacks though its efficiency is higher.

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